

Cardiac CINE MRI Semantic Segmentation Utilizing a Deep Learning Approach

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Abstract—Diagnosis of cardiac problems including Myocardial infarction damage/scarring, Cardiomyopathy, and valve diseases heavily relies on modern non-invasive magnetic resonance imagery. Analysis of CMR images is crucial for the early detection of fatal cardiac diseases. However, this task is human-intensive and requires manual inspection and delineation while being prone to human errors. Machine learning and deep learning approaches proved to offer excellent quality in medical imagery including brain and colon MRI. In this study, we propose the usage of 3 different deep-learning architectures to perform semantic segmentation of the heart's key organs with 3 different imaging planes, 4CH, 2CH and SAX. Our proposed architecture is trained and tested on 105 real-life cases of manually annotated CMR images resulting in a state-of-the-art dice score of 98% and an F1-score of 99%. Our proposed architecture can be used as a first step analysis before being input for other disease-specific models or simply to aid the medical human expert for easy decision making.

Keywords—Medical diagnostic imaging, Cardiac MRI, Medical Image Processing, Deep Learning, Semantic segmentation, Supervised learning, Hyperparameter optimization, Convolutional neural networks, U-Net, Fully-Convolutional Networks

I. INTRODUCTION

Diagnosing cardiac diseases, scars and abnormalities has been dictated by medical imagery in the past few decades. Computed Tomography (CT) scans was largely used to diagnose coronary artery diseases and for calcium scoring [1], it leverages X-rays with computer-aided reconstruction. CT scans are known for being fast and cheap but excessive radiation acts as its main downside. CT scans underwent major improvements since the utilization of deep learning to reconstruct x-ray images [2]. Echocardiography is another imaging technique that is used to assess dynamic functions of the heart by sending sound waves to create real-time moving images of the heart during blood flow. Echocardiography is used in detecting early signs of Heart Failures (HF) and Coronary Artery Diseases (CAD) as well as diagnosing patients with diabetes mellitus [3]. However, Echocardiography yields very low resolution images/frames that fail to discriminate core tissue of the heart for evaluating volume or density or area of the heart's ventricles or the myocardium or the volume of cavity of the chambers. Magnetic Resonance Imaging (MRI) was first introduced in Australia and used on a live human patient on July 3, 1977, by Dr. Raymond Damadian and his

team, who scanned a man's chest to successfully demonstrate the technology. It is a non-invasive and safe technique that can capture the core tissues of the heart for medical diagnosis of several diseases and for detecting scars and to measure the amount of cardiac fatty tissues around the heart [4]. However, MRI is slow, expensive and needs manual delineation and extensive manual efforts to segment the heart's core tissue areas which is prone to human errors and inaccuracies as well as being expensive for this human labor hours. In this work, we propose multiple deep learning based approaches to overcome the issue of manual delineation and human annotation of CMR images and also compare the results between our approach in-relation to the related work in literature. We also present a pipeline of preprocessing the raw CMR images to make them suitable for computational tasks and prevent unpredicted results while boosting explainability of our approach.

II. RELATED WORK

Cardiovascular diseases remain among the leading causes of mortality worldwide, making early diagnosis and accurate cardiac assessment essential in modern healthcare systems. Cardiac Magnetic Resonance Imaging (CMR) is considered one of the most reliable imaging modalities for evaluating cardiac anatomy, ventricular function, myocardial abnormalities, tumors, and structural heart diseases because of its high spatial resolution and superior soft-tissue contrast [19]. However, manual segmentation of cardiac structures from cine MRI images is time-consuming, labor-intensive, and highly dependent on clinical expertise. Therefore, deep learning-based automatic cardiac MRI segmentation methods have received significant attention in recent years.

Early deep learning approaches for medical image analysis mainly relied on Convolutional Neural Networks (CNNs). LeCun et al. [5] demonstrated the effectiveness of CNN architectures in hierarchical feature extraction and automatic image representation learning. Although CNN-based approaches achieved strong performance in image classification tasks, they were not originally designed for dense pixel-wise medical image segmentation, which limited their ability to accurately preserve anatomical boundaries in cardiac MRI applications.

To address these limitations, Long et al. [6] introduced Fully Convolutional Networks (FCNs), replacing fully connected

TABLE I
SUMMARY OF RELATED WORK

Paper	Technique	Findings	Limitations	Research Gaps
Gradient-Based Learning Applied to Document Recognition [5]	CNN	Effective hierarchical feature extraction and automatic image representation learning	Limited pixel-wise segmentation capability	Need for accurate medical image segmentation frameworks
Fully Convolutional Networks for Semantic Segmentation [6]	FCN	Enabled end-to-end semantic segmentation	Coarse segmentation boundaries due to repeated pooling	Preservation of fine anatomical cardiac structures
U-Net: Convolutional Networks for Biomedical Image Segmentation [7]	U-Net	Improved biomedical image segmentation accuracy using encoder-decoder architecture	Sensitive to data variability and noise	Limited multi-view cardiac MRI evaluation
UNet++: A Nested U-Net Architecture for Medical Image Segmentation [8]	UNet++	Enhanced multi-scale feature fusion and segmentation accuracy	Increased computational complexity	Limited application in multi-view cine MRI analysis
Attention U-Net: Learning Where to Look for the Pancreas [9]	Attention U-Net	Improved focus on important anatomical regions	Increased computational cost	Limited stability analysis across cardiac MRI views
Recurrent Residual Convolutional Neural Network Based on U-Net (R2U-Net) [10]	R2U-Net	Improved feature propagation and segmentation robustness	High memory consumption	Scalability challenges on large medical datasets
An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale [11]	Vision Transformer (ViT)	Captured global contextual image dependencies effectively	Requires large-scale datasets and high computation	Limited direct medical segmentation applications
TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation [12]	TransUNet	Combined CNN local features with transformer global representation	High GPU memory usage	Need for lightweight implementations in medical imaging
UNETR: Transformers for 3D Medical Image Segmentation [13]	UNETR	Strong transformer-based segmentation performance	Complex training process	Limited comparative analysis with CNN and FCN methods
Swin-Unet: Unet-Like Pure Transformer for Medical Image Segmentation [14]	Swin-Unet	Improved multi-scale contextual representation	High model complexity	Limited real-world deployment in cardiac MRI systems
Segment Anything [15]	SAM	Strong segmentation generalization capability	Requires extensive computational resources	Limited adaptation to cardiac cine MRI segmentation
MedSAM: Segment Anything in Medical Images [16]	Foundation Model	Improved medical image segmentation generalization	Requires extensive fine-tuning	Limited validation on cine cardiac MRI datasets
Automated Cardiovascular Magnetic Resonance Image Analysis with Fully Convolutional Networks [17]	FCN-Based Cardiac MRI Segmentation	Automated cardiac chamber segmentation and functional analysis	Limited multi-view evaluation	Insufficient analysis across SAX, 2CH, and 4CH views
Convolutional Neural Networks with Shape Priors for Cardiac MRI Segmentation [18]	CNN + Shape Priors	Improved anatomical consistency in cardiac segmentation	Sensitive to image variability	Limited generalization across datasets
Statistical Normalization Techniques for Magnetic Resonance Imaging [19]	MRI Normalization	Improved MRI consistency and preprocessing quality	Dataset dependency	Need for unified normalization frameworks
A Comparative Analysis of Data Augmentation Approaches for MRI Scan Images [20]	MRI Data Augmentation	Improved robustness and generalization performance	Augmentation sensitivity	Limited cardiac MRI augmentation studies
USE-Evaluator: Performance Metrics for Medical Image Segmentation Models [21]	Segmentation Evaluation Metrics	Improved reliability of segmentation evaluation	Complex evaluation process	Need for unified cardiac MRI evaluation metrics
Proposed Work	CNN + FCN + U-Net	Comparative multi-view cardiac MRI segmentation across SAX, 2CH, and 4CH views	Limited to supervised learning approaches	Future integration with transformer-based hybrid frameworks

layers with convolutional operations to enable end-to-end semantic segmentation. FCNs significantly improved dense prediction and spatial segmentation performance; however, repeated pooling and downsampling operations often resulted in coarse segmentation boundaries and loss of fine anatomical details, which are critical in cardiac MRI analysis.

A major advancement in biomedical image segmentation was achieved through the introduction of U-Net by Ronneberger et al. [7]. The U-Net architecture employed an encoder-decoder framework with skip connections that transferred high-resolution spatial features from the encoder to the decoder. This architecture substantially improved localization accuracy and became one of the most widely adopted segmentation models in medical imaging and cardiac MRI analysis.

Several extensions of U-Net were later proposed to further

improve segmentation performance. Zhou et al. [8] introduced UNet++, which utilized nested dense skip pathways to enhance multi-scale feature fusion and segmentation accuracy. Oktay et al. [9] proposed Attention U-Net, integrating attention mechanisms to improve focus on important anatomical structures while suppressing irrelevant background information. Similarly, Alom et al. [10] developed R2U-Net by combining recurrent convolutional layers with residual learning to improve feature propagation and segmentation robustness.

Transformer-based architectures have also demonstrated strong performance in computer vision and medical image analysis. Dosovitskiy et al. [11] introduced the Vision Transformer (ViT), showing that self-attention mechanisms can effectively capture long-range dependencies within images.

Building upon this concept, Chen et al. [12] proposed TransUNet, combining transformer encoders with CNN-based decoders to extract both global contextual information and local spatial features. Hatamizadeh et al. [13] further introduced UNETR, a transformer-driven segmentation framework for medical imaging applications. More recently, Cao et al. [14] proposed Swin-Unet, which utilized shifted-window transformer mechanisms to improve multi-scale contextual representation in medical image segmentation tasks.

Foundation models have recently emerged as promising approaches in medical image analysis. Kirillov et al. [15] introduced the Segment Anything Model (SAM), which demonstrated strong generalization capability across various segmentation tasks. Based on this concept, MedSAM [16] adapted SAM for medical image segmentation applications and achieved promising performance across multiple medical imaging modalities. Nevertheless, these approaches generally require substantial computational resources and extensive fine-tuning for cardiac cine MRI applications.

In cardiac MRI specifically, Bai et al. [17] proposed an FCN-based framework for automated cardiovascular magnetic resonance analysis, enabling automatic segmentation of cardiac chambers and functional assessment. Zotti et al. [18] further improved anatomical consistency by incorporating shape priors into CNN-based cardiac MRI segmentation models.

Several studies also emphasized the importance of pre-processing and normalization techniques in MRI-based deep learning systems. Shinohara et al. [19] demonstrated that MRI normalization significantly improves image consistency and segmentation performance across datasets. Safdar et al. [20] showed that data augmentation techniques can improve generalization capability and robustness in MRI image analysis. Furthermore, Ostmeier et al. [21] highlighted the importance of reliable evaluation metrics for medical image segmentation, especially in challenging annotation settings.

Although significant progress has been achieved in deep learning-based cardiac MRI segmentation, several challenges still remain. Previous studies have investigated different segmentation architectures, including CNN-, FCN-, and U-Net-based models, and some works reported comparative evaluations among these approaches. However, many existing studies focused primarily on improving segmentation accuracy on specific datasets or single cardiac MRI views, while limited attention was given to establishing a unified comparative framework across multiple cine MRI representations such as SAX, 2CH, and 4CH views. In addition, several approaches emphasized architectural complexity and performance improvements without sufficiently analyzing training consistency, model robustness, and practical applicability in multi-view cardiac MRI segmentation tasks.

Motivated by these limitations, the present work introduces a comparative multi-view cardiac MRI segmentation framework based on CNN, FCN, and U-Net architectures. Unlike previous studies that focused on a single architecture or individual MRI representation, the proposed framework evaluates segmentation performance consistently across SAX,

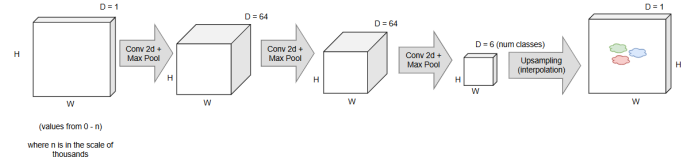


Fig. 1. CNN with a final upsampling layer architecture

2CH, and 4CH cine MRI views using unified preprocessing, training, and evaluation procedures. The proposed study aims to provide a balanced comparative analysis of segmentation accuracy, robustness, and practical applicability for automated cardiac MRI segmentation.

III. METHODOLOGY

A. CNN Architecture

1) *Description:* We implemented a simple raw CNN architecture as a baseline for our work and to prove that raw CNN in its traditional form is not the most suitable for segmentation tasks. We prepared a series of convolutional layers followed by Max-Pooling layers chained together 3 times, then routed to a final upsampling layer using interpolation to reconstruct the mask with correct dimensions. The full architecture is shown in figure 1

2) *Network Implementation:* Since we targeted the usage of the MICCAI dataset which originally had 105 cases, the data is input to our preprocessing pipeline discussed later in this paper. A network consisting of 3 Conv layers with Max Pools in-between was used. The first Conv layer had 1 input channel and 64 output channels with a trainable kernel size of 3x3 and 1-to-1 padding to prevent dimensionality issues. This layer expands the depth of the initial image to 64. This layer is followed by a Max-pooling layer with kernel size 2x2 and a stride of 2, hence, decreasing the width and height of the first intermediate representation. The second Conv layer accepts a depth of 64 and outputs the same depth of 64 with trainable kernels of size 3x3 and 1-to-1 padding. This layer is also followed by a Max-Pooling layer of size 2x2 and a stride length of 2. The final Conv layer accepts a depth of 64 but outputs a depth of num_classes where the number of classes depends on our segmentation task (4CH, 2CH or SAX), with the same kernel size and padding. This final layer is not followed by a max pooling layer, but instead, it is followed by an interpolation layer to restore the original width and height of the image, sort of like expanding the image after downscaling it due to the conv and max pooling layers, using a bilinear interpolator.

Data is split into a training, validation and testing set. During training, data is input to the preprocessing pipeline then gets convolved and max-pooled 2 consecutive times causing its dimensions to change noticeably (its width and height will decrease significantly while its depth increases), the final convolution layer is activated then followed by an interpolation process to retrieve the original dimensions of the image.

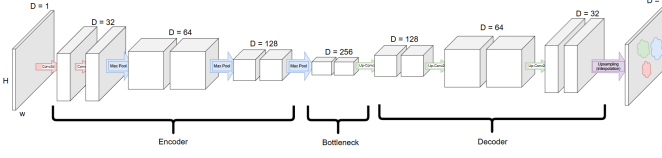


Fig. 2. Fully Forward Convolution Network without Skip Connections, consisting of an encoder, a decoder and an intermediate bottleneck.

This approach is clearly the worst of other alternatives. Because the image's details are compiled and combined together till it reaches a dense block of features with small dimensions. Using this dense block to infer classification (per-pixel) on the source input image (which has much larger resolution and different dimensions) is not correct. This dense block of features represent the entire meaningful features of the image as a whole not per-pixel. deep features can be obtained when going deeper, spatial location information is also lost when going deeper We need to somehow preserve the pixels-data from the original image (maybe by using skip-connections) or to upsample this dense block of features (using a decoder) to correlate the features in the dense block back to the original pixels and to finally create a mask.

B. Fully Convolutional Network (FCN) Architecture

One layer of immediate interpolation from low resolution image (after applying multiple convolutions and max-pooling) to the original dimensions is subject to heavy errors. It is a better approach to make a dedicated decoder network with trainable parameters. The Fully Convolutional Network (FCN) achieves this goal by utilizing an encoder-decoder architecture, where the encoder is simply a CNN while the decoder is a series of De-Convolutional layers (sometimes called Up-Convolution or Reverse Convolution) with trainable features as much as those inside the CNN. The encoder's job is to try to understand the complex relationships between pixels and spatial data of the image, while the decoder's job is to try to relate which pixels belong to which class, given the relationship information provided by the encoder. It is an update on the pure CNN approach by adding a simple decoder.

1) *Network Implementation:* Our FCN approach used 3 encoder stages, a bottleneck stage and 3 decoder stages. All stages are sequential without any skip connections, as we will introduce updates on this architecture later. Each stage in the encoder consists of 2 convolution layers followed by a max-pooling layer. The bottleneck stage acts as a pipe to the decoder, it consists of 2 convolution layers back to back. Each stage in the decoder consists of 1 Up-Convolution layer followed by 2 convolution layers, the last ever convolution layer maps to the number of classes before being input to a softmax layer. For all convolution and deconvolution layers, their output are subject to a RELU activation function to introduce non-linearity and increase the complexity of the model. An accurate representation of this approach is shown in figure 2

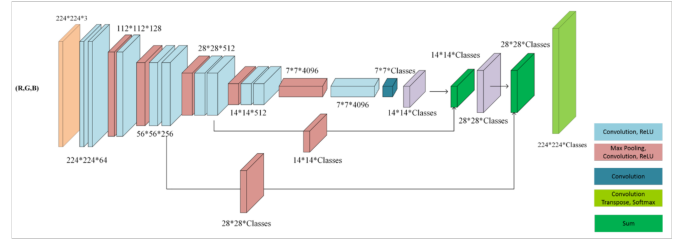


Fig. 3. FCN with chosen skip-connections architecture

C. FCN with Skip Connections Architecture

Since our main goal in this problem is not to create images as in generative AI, we want to make use of the original image in order to annotate some pixels inside it. Preserving the features from the original image is crucial as we are not creating a whole new non-existing image. We implemented some skip-connections where we take a copy of the image after some certain layers in the encoder and simply concatenate this image to its counterpart in the decoder hence, preserving some of the original image features to be utilized during the up-convolution phases. Those skip-connections are not totally from each encoder phase to each decoder phase as the U-Net's architecture, but from some chosen layers in the encoder to some chosen layers in the decoder. The general architecture of this approach is shown in figure 3 The update is that we implemented 2 skip-connections from the third and fourth stage in the encoder, to the second and third stage in the decoder respectively, using a summation operation since both images (skipped from the encoder to its counterpart in the decoder) have the same dimensions. Training this type of network took longer time and required more processing resources (GPU, RAM and CPU utilization). However, its results were already explainable and robust.

D. U-Net Architecture

To achieve accurate and reliable cardiac MRI segmentation, a U-Net-based deep learning architecture was implemented in this study due to its effectiveness in biomedical image segmentation tasks and its ability to preserve both semantic and spatial information [7]. The adopted model was designed to segment cardiac cine MRI images obtained from multiple cardiac views, including Short-Axis (SAX), Two-Chamber (2CH), and Four-Chamber (4CH) representations.

Figure 4 presents the detailed architecture of the implemented U-Net framework used in this work.

The implemented U-Net follows a symmetric encoder-decoder architecture composed of three major components: the encoder (contracting path), the bottleneck layer, and the decoder (expanding path). The overall structure was specifically designed to balance contextual feature extraction and accurate pixel-level localization, which are essential requirements in cardiac MRI segmentation tasks.

The input to the network consists of grayscale cardiac MRI slices resized to 256×256 pixels. Prior to training, all

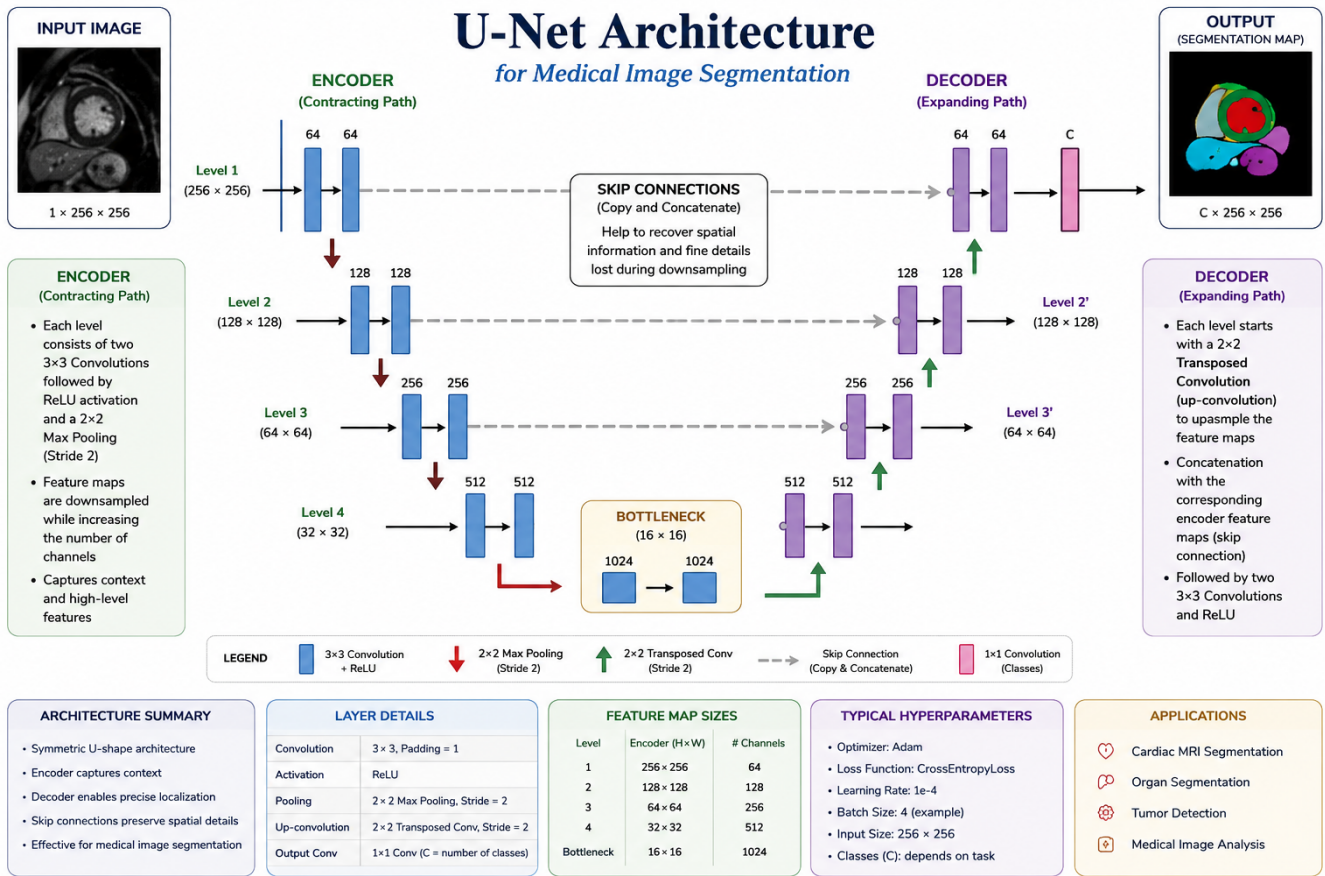


Fig. 4. Detailed architecture of the implemented U-Net model for cardiac MRI segmentation. The network consists of an encoder path for hierarchical feature extraction, a bottleneck layer for deep semantic representation, and a decoder path for spatial reconstruction. Skip connections are employed to preserve fine anatomical details and improve localization accuracy during segmentation.

MRI images were normalized to improve intensity consistency across different scans and patients. This preprocessing step reduces variability caused by imaging conditions and enhances training stability.

The encoder path is responsible for extracting hierarchical spatial and semantic features from the input MRI images. Each encoder level contains two consecutive convolutional layers with kernel size 3×3 , followed by Rectified Linear Unit (ReLU) activation functions. Padding was applied to preserve spatial dimensions after convolution operations. A 2×2 max-pooling layer with stride 2 was then used to progressively reduce spatial resolution while increasing the receptive field of the network.

As illustrated in Fig. 4, the number of feature channels increased gradually through the encoder levels from 64 to 128, 256, and 512 feature maps. This progressive increase enables the model to learn low-level image characteristics such as edges and textures in earlier layers while capturing more complex anatomical structures and contextual cardiac information in deeper layers.

At the deepest stage of the network, referred to as the bottleneck layer, the extracted feature maps reached a spatial size of 16×16 with 1024 feature channels. This layer provides

a compact semantic representation of the cardiac structures and surrounding tissues. The bottleneck stage plays an important role in capturing high-level contextual information required for accurate segmentation of ventricular cavities, myocardium, and cardiac boundaries.

The decoder path reconstructs the segmentation mask by progressively restoring the spatial resolution of the feature maps. Each decoder level begins with a 2×2 transposed convolution (up-convolution) operation used for upsampling the feature maps. The upsampled features are then concatenated with the corresponding encoder features through skip connections.

The skip connections represent one of the most important components of the U-Net architecture. During the encoder downsampling process, some fine spatial details may be lost due to repeated pooling operations. Therefore, skip connections directly transfer high-resolution encoder features to the decoder layers, allowing the network to recover anatomical boundaries and preserve fine structural information. This mechanism significantly improves localization accuracy and enhances segmentation quality, particularly in medical imaging applications where precise boundary delineation is critical.

After concatenation, two additional 3×3 convolutional

layers followed by ReLU activation functions are applied to refine the reconstructed feature maps. The decoder gradually restores the original spatial resolution while reducing the number of feature channels until the final segmentation map is generated.

At the final stage of the network, a 1×1 convolutional layer is used to project the extracted features into pixel-wise class probabilities corresponding to the target cardiac structures. The output segmentation mask preserves the same spatial dimensions as the original input image.

The implemented model was trained using the Adam optimization algorithm with a learning rate of 10^{-4} . Cross-Entropy Loss was employed as the primary optimization objective for pixel-wise classification. In addition, mixed precision training (Automatic Mixed Precision, AMP) was utilized to reduce GPU memory consumption and improve computational efficiency during training. Model checkpoints were continuously saved throughout the training process to preserve the best-performing model weights according to validation performance.

The proposed U-Net framework demonstrated strong capability in extracting multi-scale cardiac features while preserving fine anatomical structures across different cardiac MRI views. The combination of encoder-decoder learning, skip connections, and hierarchical feature extraction enabled the model to generate accurate and robust segmentation masks suitable for automated cardiac MRI analysis.

IV. DESIGNING THE LOSS FUNCTION

Since the dice loss itself does not have the ability to differentiate or assign weights or penalties for different classes, and since the images were subject to some perplexity in the background (some background pixels get wrongly masked). We would want to penalize the model when it wrongly masks pixels that originally belong to certain classes. Cross Entropy loss function has the option to be updated to make use of a weight tensor where each element in this tensor constitutes a penalty weight for misclassifying pixels belonging to such a class. Hence, we propose the usage of a combined loss function that is a summation of both the Dice loss and the weighted Cross Entropy where we can freely control the weight tensor to penalize the model according to our needs.

Combined Dice Loss and Weighted Cross-Entropy Loss

Softmax Probabilities:

$$p_{i,c} = \frac{e^{z_{i,c}}}{\sum_{k=1}^C e^{z_{i,k}}} \quad (1)$$

Weighted Cross-Entropy Loss (per pixel):

$$\mathcal{L}_{\text{WCE}} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \log(p_{i,y_i}) \quad (2)$$

Dice Loss (excluding background class $c = 0$):

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{N} \sum_{i=1}^N \frac{1}{C-1} \sum_{c=1}^{C-1} \frac{2 \sum_{j=1}^{HW} p_{i,c,j} y_{i,c,j} + \epsilon}{\sum_{j=1}^{HW} p_{i,c,j} + \sum_{j=1}^{HW} y_{i,c,j} + \epsilon} \quad (3)$$

Combined Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{WCE}} + \mathcal{L}_{\text{Dice}} \quad (4)$$

Definitions:

- N : number of samples (batch size)
- C : number of classes
- H, W : spatial dimensions
- w_c : weight for class c
- y_i : ground-truth class label at pixel i
- $y_{i,c,j}$: one-hot encoded ground truth
- $p_{i,c,j}$: predicted probability
- ϵ : smoothing constant

V. DATASET PREPARATION

The dataset is collected and carefully annotated by cardiologists and radiologists from Beijing Anzhen Hospital & collaborative clinical centers[22]. The dataset consists of two collections of images, CINE and LGE, each collection for a special type of problem, the LGE type of imagery is out of scope of this work. The CINE image collection consists of 105 cases -patients- with 3 views per patient which are 4 chamber view (4CH), 2 chamber view (2CH) and short axis view (SAX). Each case has its own annotation files. Each file comes in its original NIfTI format which is the standard image format for MRI in the medical industry, dealing with this type of images programmatically was a challenge. Other challenges will be addressed, as treating NIfTI images as if they were pure PNG or bitmap files might yield biased results.

A. Data Augmentation

Since the provided dataset is not enough for computationally-intensive deep learning models to achieve good results and prevent overfitting, we propose the usage of extra data and annotate it automatically using another model. However, the usage of augmented data should be minimal compared to the size of the original dataset so as not to introduce artificial artifacts in the new combined dataset and affect the overall performance of our model. Meta AI introduced the Segment Anything Model (SAM) in 2023 [23] which is a pretrained semantic segmentation model that is strengthened by the prompting capabilities to improve the localization of pixels, it is widely used in dynamic MRI and medical imagery for its quick results and ease of use. We propose the usage of SAM to annotate an externally collected small set of CINE MRI images to aid the training of our main deep learning model. The portion of augmented data comprises 10% of all used data before splitting into training, testing and validation. The usage of SAM stemmed from its support of prompt engine where we can manually include or exclude parts of each area belonging to a certain class unlike the YOLOv8-dependent approach by Muhammad Farhan et al. in their review of data augmentation [24] which was limited to the capabilities of the pretrained YOLOv8 model to segment parts inside MRI images without being fine-tuned with this type of imagery 5.

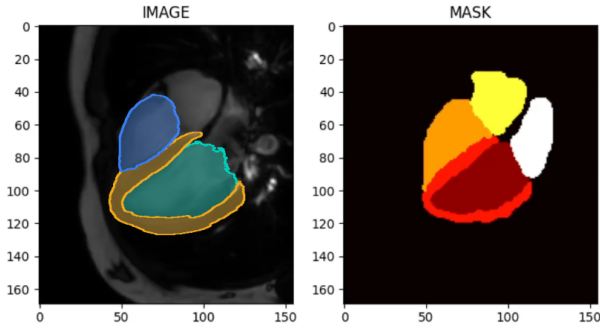


Fig. 5. Using SAM2 during Data Augmentation

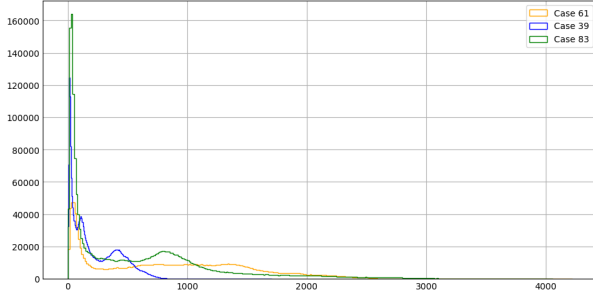


Fig. 6. Different MRI cameras output different histograms with different ranges of values

B. Dataset Exploratory Analysis

Since the NIfTI format should not be treated like the average png or bitmaps, each pixel's value range should be inferred from different images, it is found that each NIfTI image has a pixel value ranging from 0 to thousands of arbitrary signal strength units where the intensity for bones is greater than the intensity of fat, and the intensity of fat is greater than the intensity of muscles. Sample histograms of randomly selected CMR images from the dataset is shown in figure 6. It is also clear that the max value of images is not 255 nor 65535 nor other max value that is compatible with digital 64-bit storage matrices. We determined that the max value in CMR images follow a certain distribution shown in figure 7. It is found that the max values range from 601 to 7428 acting as a challenge when we input those values into a deep-learning model for training and inference. Those large values may break the model and assign unreal priority to those high value pixels than pixels from other images although both pixels govern the same type of organic tissue.

C. Scaling Values

Values should be scaled to prevent breaking any statistical model. Hence, we tried different types of scalers which are 1- MinMax scaler, 2- Z-Score Scaler and 3- RobustScaler. The MinMax scaler is a simple normalizer from $[ABS_MIN - ABS_MAX]$ to $[0 - 1]$ but requires the looping on all images to infer the absolute maximum and absolute minimum pixel values. The MinMax scaler is proven to be the most reliable and robust to high variant max values as discussed previously

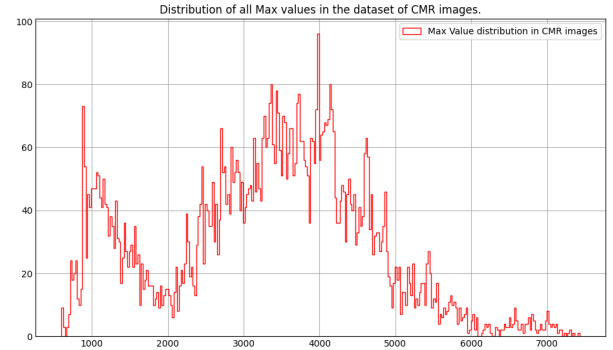


Fig. 7. Distribution of all Max values in CMR images in the dataset.

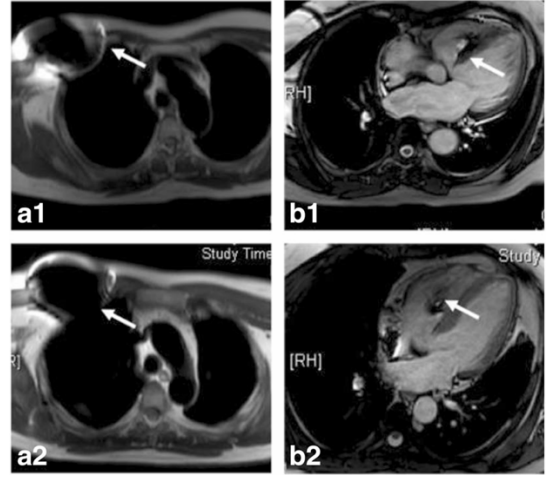


Fig. 8. Very dark spots in CMR due to installation of a pace maker

and the scaling is still relative meaning that each part preserves its relative brightness with other regions in the image. After deep analysis, we resorted to the adoption of the local min-max scaler. This is because an outlier image may be input from another camera vendor that has a max value bigger than our computed global ABS_MAX value and such image values will break our model during inference or even during training a global min-max scaler. Side cases can be introduced as stated in [25] such as a case when there is a small piece of metal in the MRI image causing very dark spots or silicon implants causing very bright spots as shown in figures 8 and 9.

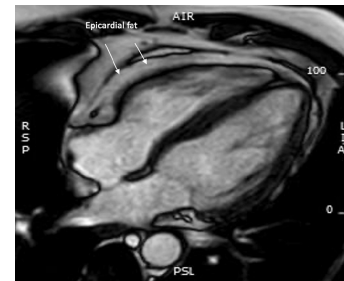


Fig. 9. Fat tissues and excessive water resulting in bright areas in CMR.

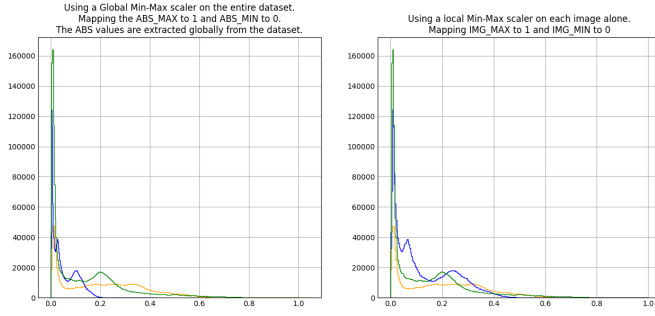


Fig. 10. Global vs Local MinMax scalers and showing histogram mismatch between different MRI cameras

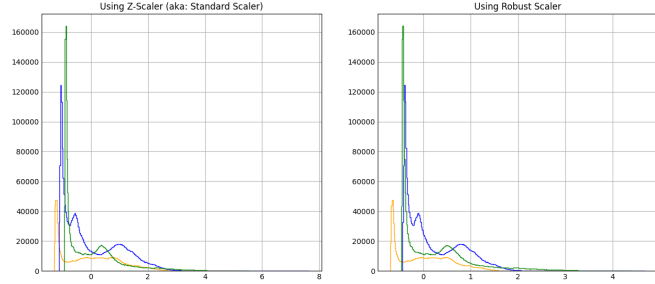


Fig. 11. Comparison between the usage of Robust Scaler and Standard z-Scaler still showing the problem of histogram mismatches.

Generally speaking, using the local min-max scaler is more reliable in normal cases than the global min-max scaler, however if a certain anomalous case as a case containing a metal piece, this will cause the entire classification of this anomalous case to be affected negatively. It is a trade off and we will accept the risk of misclassify a single anomalous case than to poison the scaling of our entire dataset from individual anomalous cases. However, the problem of histogram mismatch between MRI cameras is still present and clearly shown in figure 10.

Other scalers were considered such as the Standard -ZScore-Scaler and Robust Scaler as used in Brain MRI normalization in Shinohara et al. study [26]. However, this approach seems not suitable for CMR images, a quick comparison is shown in figure 11 which proves that Z-scaler and robust scaling images of the dataset can bring histogram mismatches and did not correctly solve the problem of histogram mismatches.

Another thing should be considered in minds is that when we use a min-max scaler, due to the very large value of the max value pixel, some features recorded in the early tail of the histogram gets squeezed together after scaling, features became very close to each other to preserve relative distance between the small values to the big values. To address this issue, clipping is considered and tested. By clipping the upper 99 percentile of each image to deter any outliers then perform scaling, this experiment's results is recorded as per figure 12.

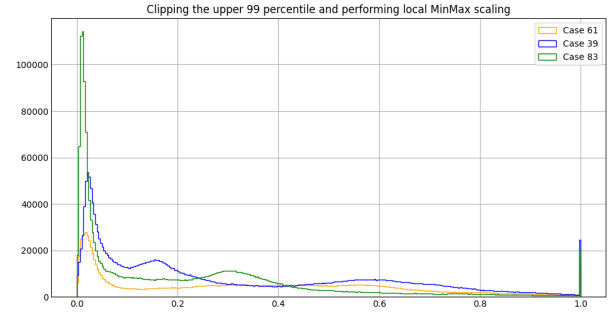


Fig. 12. Clipping 99 percentile and MinMax scaling.

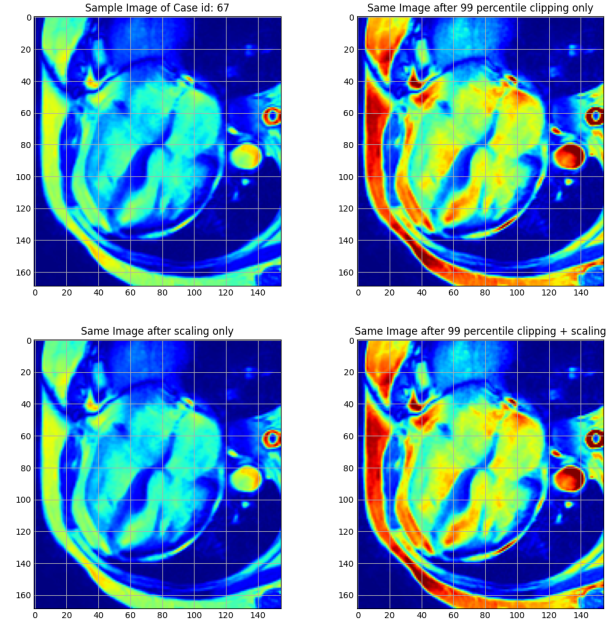


Fig. 13. Comparing the effect of 99% Clipping and MinMax scaling and mixing both operations together

now with any abnormalities normalized due to the clipping operation.

Final human checking of those images after clipping and after MinMax scaling shows a great shift in the coloring scheme of the MRIs which can alter the human detection/diagnosis of images, this great alteration shown in figure 13 can perplex humans and hence can perplex statistical deep learning models from grabbing the features that humans look for during their daily checkup. The visualization color map was set to "jet" as it captures the slight differences of pixel values in an excellent manner better than the grey-scale visualization to the naked eye.

VI. MEASUREMENT METRICS

Semantic segmentation of images is not effectively measured by the F1 accuracy as overall image classification tasks, as some side cases where the predicted mask does a poor job overlapping with the ground truth mask but the overall f1-score reflect satisfactory numbers, side cases like these

prove the insuitability of using the F1-score for measuring the performance of semantic segmentation. Since we greatly care about maximizing overlaps between the ground-truth masks and out predicted masks, another metric dictates our measurement which measures this overlap effectively. Dice and intersection over union -IoU- scores (sometimes referred to as Jaccard Index) are utilized as our measurement metrics along with using plain f1 score normally. Formula for computing the Dice score and the IoU scores are shown by:

$$\text{Dice Score}(X, Y) = \frac{2|X \cap Y|}{|X| + |Y|}$$

$$\text{IoU}(X, Y) = \frac{|X \cap Y|}{|X \cup Y|}$$

Dice score is used to emphasize the importance of the mask more than background pixels. It is found that Dice score is mostly used in medical MRI segmentation as stated in [27]. However we are going to record the IoU and also the F1-score of our approaches.

VII. RESULTS AND EVALUATION

A. Evaluating Pure CNN

Pure CNN is clearly not the best approach for a semantic segmentation task, this is shown and proven with the following results.

TABLE II
CNN TRAINING LOSS ACROSS 50 EPOCHS

Epoch	Loss	Epoch	Loss
1	2.4875	26	1.9047
2	2.3411	27	1.8979
3	2.2484	28	1.8913
4	2.1894	29	1.8849
5	2.1485	30	1.8785
6	2.1194	31	1.8725
7	2.0972	32	1.8667
8	2.0792	33	1.8612
9	2.0640	34	1.8558
10	2.0504	35	1.8506
11	2.0380	36	1.8455
12	2.0264	37	1.8406
13	2.0154	38	1.8358
14	2.0051	39	1.8311
15	1.9952	40	1.8266
16	1.9856	41	1.8221
17	1.9764	42	1.8178
18	1.9674	43	1.8135
19	1.9587	44	1.8094
20	1.9503	45	1.8054
21	1.9422	46	1.8015
22	1.9342	47	1.7976
23	1.9264	48	1.7939
24	1.9190	49	1.7903
25	1.9117	50	1.7867

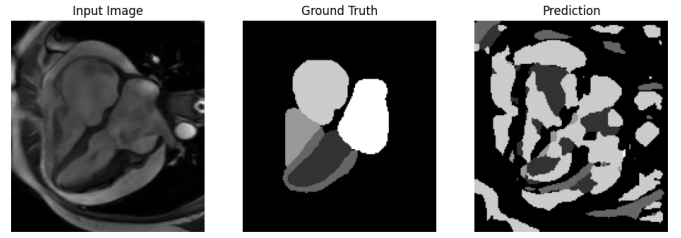


Fig. 14. Using the Pure CNN model for inference after training for 50 epochs for the 4CH segmentation problem

TABLE III
CNN CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
0	0.87	0.83	0.85	49,703,651
1	0.54	0.06	0.11	4,722,806
2	0.05	0.03	0.04	2,918,103
3	0.27	0.02	0.04	2,687,392
4	0.11	0.79	0.20	2,183,769
5	0.00	0.00	0.00	2,878,854
Accuracy	0.67			65,094,575
Macro Avg	0.31	0.29	0.21	65,094,575
Weighted Avg	0.72	0.67	0.67	65,094,575

TABLE IV
CNN ARCHITECTURE CONFIGURATION

Layer	Configuration	Output Channels
Conv1	3 × 3 Kernel, Padding=1	64
MaxPool1	2 × 2 Pooling, Stride=2	64
Conv2	3 × 3 Kernel, Padding=1	64
MaxPool2	2 × 2 Pooling, Stride=2	64
Conv3	3 × 3 Kernel, Padding=1	6

After extracting this output, the Pure CNN approach is discontinued from our tests as it was proved not suitable for further trial and analysis. We have saved the model and used it for inference on a sample from our testing set and the results were pretty away from the ground truth as in figure 14

B. Evaluating the FCN

The fully convolutional network (FCN) was used according to the setup in section III-C. The network is trained for 75 epochs using a Cosine Annealing learning rate scheduler with starting lr=1e-4 till lr=1e-6. For the model to focus on the core tissue rather than the background, a loss weight of 0.3 was assigned to the background pixels while 1.0 loss weight was assigned to all other classes regardless of their number (6 classes in 4CH problem, 4 classes in 2CH problem and 4 classes in SAX problem).

1) *Evaluating the FCN on the 4CH problem:* There is a noticeable jump of training and validation accuracies (and dice scores) when the learning rate scheduler was used rather than using a fixed learning rate of 0.0001 as in the Pure CNN approach. The learning curve showing the training and validation accuracy, F1-score and dice score along with the respective losses is shown in the following series of figures

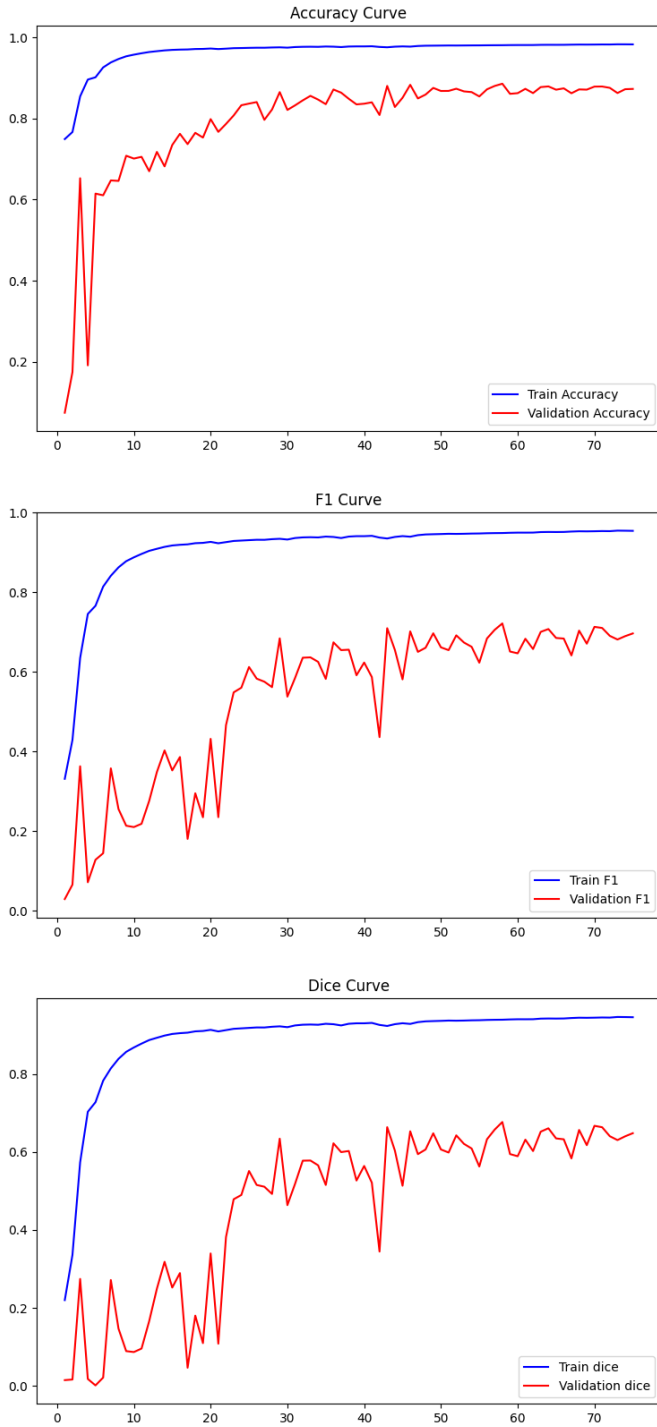


Fig. 15. Training and validation accuracy, F1-score and dice score along with the respective losses during training the FCN on the 4CH problem's dataset.

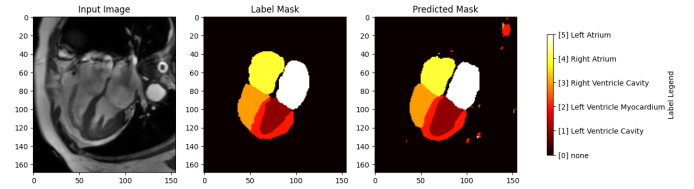


Fig. 16. A predicted mask sample of using the FCN model for inference on the 4CH problem.

From the figures 15, we concluded that the model showed a fluctuating dice score possibly due to the usage of the cosine annealing scheduler where the learning rate does not constantly decrease nor constantly increase but fluctuates between increasing and decreasing to overcome overfitting or getting stuck in a local minima during training. This model is tested on the test set comprising of 30% of the original dataset length and the results were already promising and shown in table V. The model is used for inference for a final human check and a sample is shown in figure 16

TABLE V
TEST CLASSIFICATION REPORT FOR FCN ON THE 4CH PROBLEM

Class	Precision	Recall	F1-Score	Support
0	0.93	0.96	0.95	24,793,216
1	0.81	0.81	0.81	2,294,082
2	0.61	0.68	0.64	1,420,128
3	0.80	0.57	0.66	1,403,608
4	0.77	0.64	0.70	1,118,066
5	0.85	0.63	0.73	1,505,090
Accuracy	0.8975			32,534,190
Macro Avg	0.79	0.71	0.75	32,534,190
Weighted Avg	0.90	0.90	0.89	32,534,190

Metric	Value
Dice Score	0.7065
Accuracy	0.8975
F1 Score	0.7470

2) *Evaluating the FCN on the 2CH problem:* The same model was also used to train on the 2CH CMR images and evaluated using the same parameters yielding the learning curves in figure 17. From the figure 17 we conclude that the validation dice score has reached a maxima then began to slowly decline indicating signs of overfitting. The symptom of fluctuating accuracies and score indicate that the scheduler is doing fine but needs tuning due to making very big jumps. The 2CH problem has 3 labels only making it computationally simpler to train than the 4CH problem and that is one of the reasons why the model seems to overfit by studying the learning curves. The overall model is tested on the 2CH test set and yielded the results shown in table VI. The model is finally used for a human check after inference on a test sample, this sample is shown in figure 18 indicating perplexity between the background and some other classes, perhaps this problem can be solved by introducing some image post-processing, making use of the fact that the heart's outer walls (Left Ventricle's Myocardium) has a round-like shape, maybe

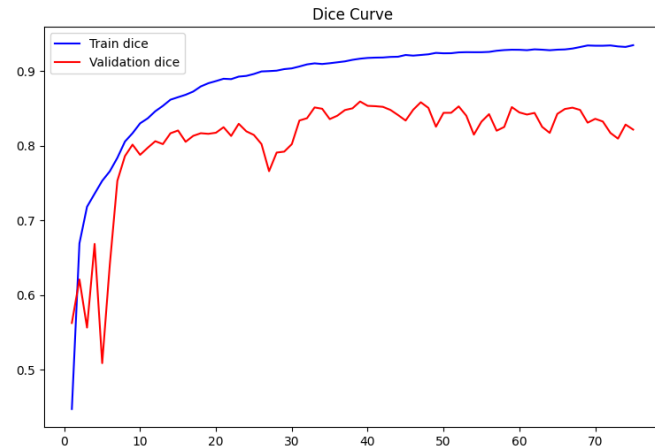
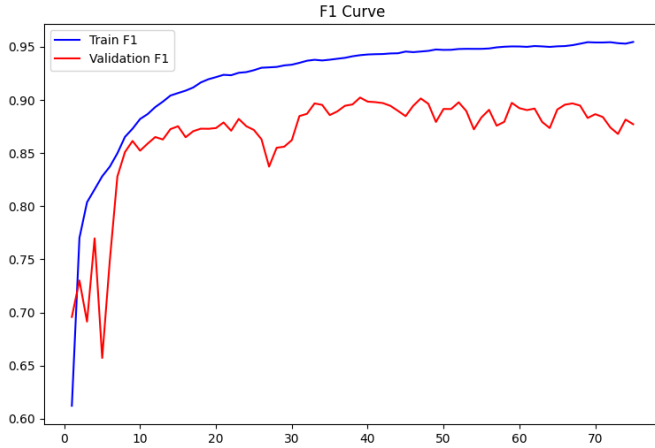
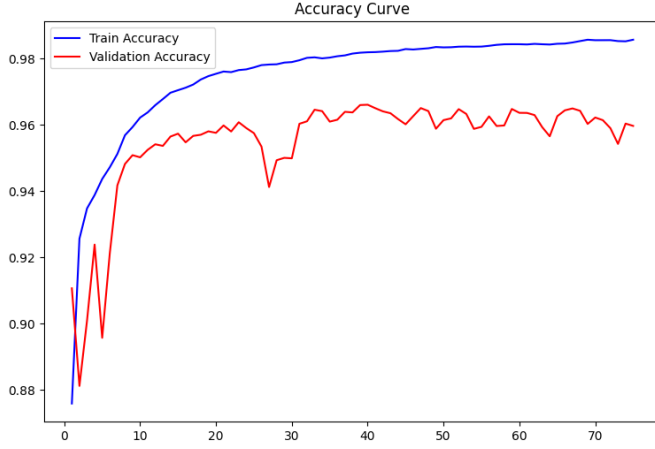


Fig. 17. Training and validation Accuracy, F1 and dice score of the FCN during training on 2CH CMR images for 75 epochs.

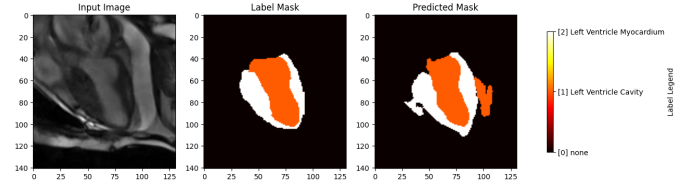


Fig. 18. Testing FCN for 2CH CMR inference, showing signs of overfitting and misclassifying some pixels belonging to the background (showing spikes).

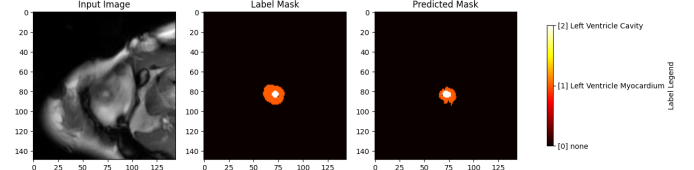


Fig. 19. Testing FCN on a distorted SAX CMR image.

some post-processing smoothing can achieve better results, this approach is left as our future work.

TABLE VI
TEST CLASSIFICATION REPORT OF THE FCN ON 2CH CMR IMAGES

Class	Precision	Recall	F1-Score	Support
0	0.99	0.98	0.98	38,781,623
1	0.84	0.90	0.87	4,898,610
2	0.76	0.72	0.74	3,697,882
Accuracy	0.9541			47,378,115
Macro Avg	0.86	0.87	0.87	47,378,115
Weighted Avg	0.95	0.95	0.95	47,378,115

Metric	Value
Dice Score	0.8056
Accuracy	0.9541
F1 Score	0.8654

3) *Evaluating the FCN on the SAX problem:* The same FCN architecture was used to train on the SAX CMR dataset and was tested using same split proportions. The SAX CMR images dataset had a small problem, the pixels to be masked are relatively small as shown in figure 19 showing the inference of the FCN on a random sample from the SAX test set. Another problem is that any small movement of the patient will affect the quality of the CMR image due to continuous motion of the patient during breathing.

The learning curve shown in 20 indicates some stability in-comparison with the learning curve of FCN on 4CH and 2CH problems.

The testing scores were shown in table VII

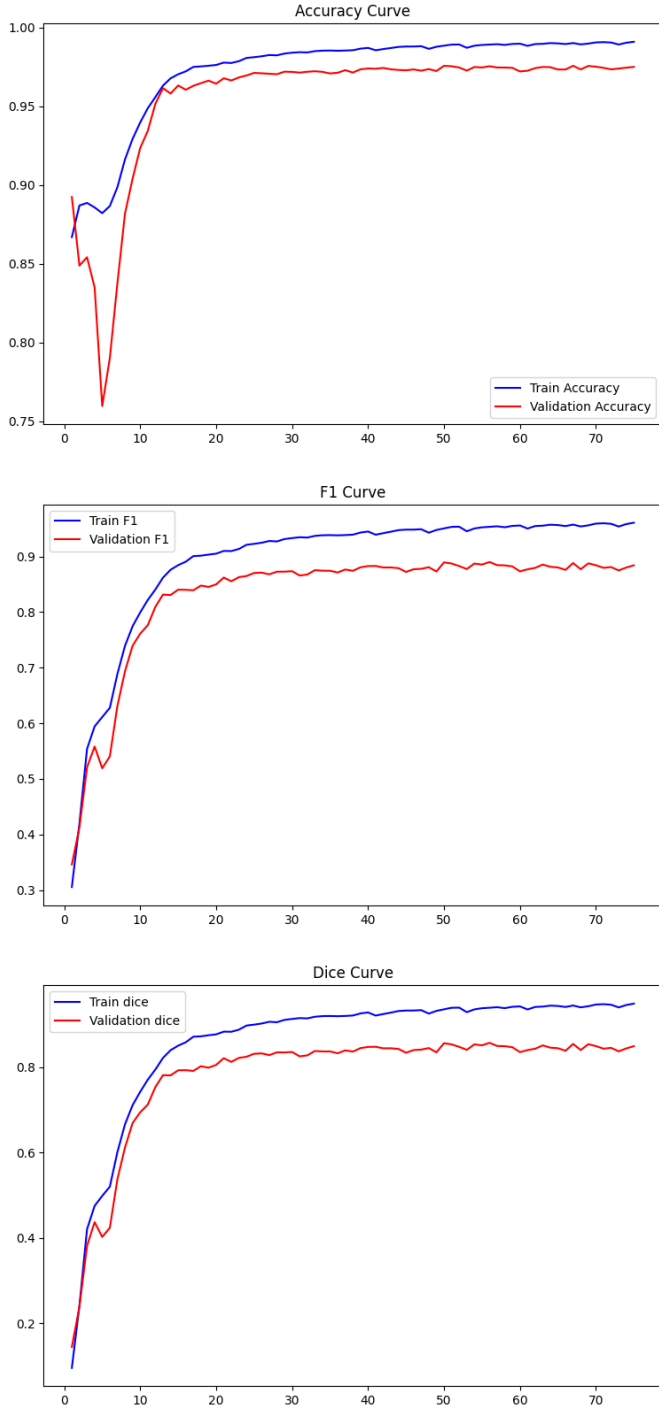


Fig. 20. Training and validation Accuracy, F1 and dice score of the FCN during training on SAX CMR images for 75 epochs, showing stability in training and validation.

TABLE VII
TEST CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
0	0.98	0.99	0.99	168,196,219
1	0.89	0.82	0.85	10,108,535
2	0.93	0.92	0.92	6,901,735
3	0.81	0.67	0.73	5,344,247
Accuracy	0.9714			190,550,736
Macro Avg	0.90	0.85	0.87	190,550,736
Weighted Avg	0.97	0.97	0.97	190,550,736

Metric	Value
Dice Score	0.8370
Accuracy	0.9714
F1 Score	0.8745

C. Evaluating the U-Net Architecture

To evaluate the segmentation performance of the proposed U-Net architecture, experiments were conducted using three different cardiac cine MRI views: the four-chamber view (4CH), the two-chamber view (2CH), and the short-axis view (SAX). These views were selected because they provide complementary anatomical information about the cardiac chambers and surrounding myocardial structures.

For all experiments, the dataset was divided into training, validation, and testing sets to ensure reliable and unbiased evaluation of the segmentation model. In the 4CH dataset, a total of 8285 MRI slices were used, including 4556 slices for training, 1242 slices for validation, and 2487 slices for testing. The same preprocessing pipeline and experimental configuration were also applied to the 2CH and SAX datasets to maintain consistency throughout the evaluation process.

The training set was used to optimize the model parameters, while the validation set was used to monitor model convergence and reduce the possibility of overfitting during training. The testing set was finally used to evaluate the generalization capability of the trained U-Net model on previously unseen cardiac MRI images.

Several quantitative evaluation metrics were used to assess segmentation performance, including segmentation loss, Dice score, accuracy, precision, recall, and F1-score. In addition, qualitative visualization analysis was performed by comparing the predicted segmentation masks with the corresponding ground-truth annotations.

The following subsections present the experimental results obtained from the 4CH, 2CH, and SAX cardiac MRI datasets.

1) *Evaluation on 4CH Cardiac MRI Dataset:* This subsection presents the experimental evaluation of the proposed U-Net architecture on the 4CH cardiac cine MRI dataset. The main objective of this experiment was to evaluate the ability of the model to accurately segment different cardiac anatomical structures from four-chamber MRI images.

The model was trained for 10 epochs using the Adam optimizer together with the Cross-Entropy loss function.

To better analyze the learning behavior of the network during training, selected epochs were used to summarize the segmentation performance of the model. Table VIII presents

TABLE VIII
TRAINING, VALIDATION, AND TESTING PERFORMANCE ON THE 4CH DATASET

Epoch	Loss			Dice Score			Accuracy		
	Train	Validation	Test	Train	Validation	Test	Train	Validation	Test
1	0.4582	0.2255	0.2218	0.8312	0.9181	0.9188	0.8312	0.9181	0.9188
5	0.0696	0.0685	0.0686	0.9714	0.9717	0.9717	0.9714	0.9717	0.9717
10	0.0531	0.0551	0.0554	0.9777	0.9771	0.9770	0.9777	0.9771	0.9770

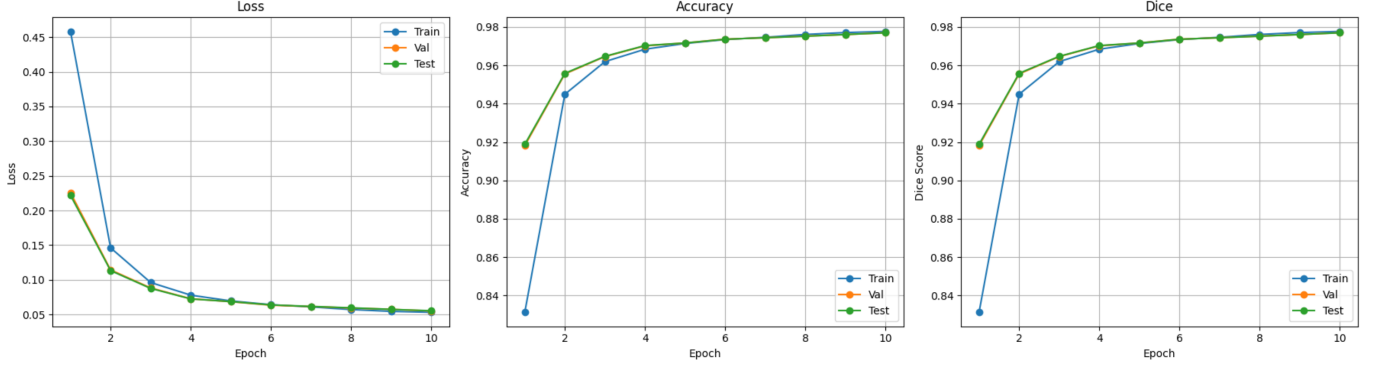


Fig. 21. Training performance curves of the proposed U-Net model on the 4CH cardiac MRI dataset. The figure illustrates the improvement in Dice score and segmentation accuracy together with the gradual reduction in segmentation loss during the training process.

the obtained loss, Dice score, and segmentation accuracy values at epochs 1, 5, and 10.

The obtained results demonstrate that the proposed U-Net model achieved stable learning performance throughout the training process. At the beginning of training, the model achieved a Dice score of 0.9188 on the testing dataset. As the training process continued, the segmentation performance improved steadily, reaching a Dice score of 0.9717 at epoch 5 and achieving the best performance at epoch 10 with a Dice score of 0.9770.

At the same time, the segmentation loss gradually decreased from 0.2218 to 0.0554, indicating that the model successfully learned meaningful spatial and contextual information from the cardiac MRI images.

Figure 21 illustrates the training curves obtained during the optimization process. The figure presents the evolution of segmentation loss, Dice score, and segmentation accuracy across the training epochs.

As shown in Figure 21, the model exhibited smooth convergence behavior during training without significant fluctuations or instability. The close similarity between the validation and testing results also indicates that the proposed framework achieved good generalization capability without severe over-fitting.

To further evaluate the segmentation quality across different cardiac structures, precision, recall, and F1-score metrics were calculated for each segmentation class. Table IX summarizes the obtained segmentation performance on the testing dataset.

The per-class evaluation results demonstrate that the proposed segmentation framework was able to accurately identify multiple cardiac anatomical structures within the 4CH MRI images. Most segmentation classes achieved high precision

and recall values, reflecting accurate feature extraction and reliable anatomical boundary localization.

Among all segmented structures, the LV cavity and left atrium achieved particularly strong segmentation performance with F1-scores of 0.96 and 0.95, respectively. Although the LV myocardium class achieved slightly lower performance compared with the remaining classes, the obtained results are still considered satisfactory because myocardial boundaries are relatively thin and more difficult to segment accurately in cine cardiac MRI images.

Figure 22 presents additional visualization examples generated by the proposed U-Net architecture on the 4CH cardiac MRI dataset.

The additional visualization results further confirm the robustness of the proposed segmentation framework and demonstrate the capability of the model to preserve fine cardiac anatomical details across different MRI samples.

2) *Evaluation on 2CH Cardiac MRI Dataset:* This subsection presents the experimental evaluation of the proposed U-Net architecture on the 2CH cardiac cine MRI dataset. The purpose of this experiment was to evaluate the capability of the model to segment important cardiac anatomical structures from two-chamber MRI images.

The model was trained for 10 epochs using the same preprocessing pipeline, optimization settings, and training configuration applied in the previous experiments.

To summarize the learning behavior of the proposed U-Net framework during training, selected epochs were used for performance comparison. Table X presents the obtained loss, Dice score, and segmentation accuracy values at epochs 1, 5, and 10.

The obtained results demonstrate that the proposed U-Net

TABLE IX
PER-CLASS SEGMENTATION PERFORMANCE ON THE 4CH DATASET

Class	Precision	Recall	F1-Score
Background	0.99	0.99	0.99
LV Cavity	0.95	0.96	0.96
LV Myocardium	0.87	0.89	0.88
RV Cavity	0.94	0.93	0.93
Right Atrium	0.94	0.92	0.93
Left Atrium	0.94	0.95	0.95
Macro Average	0.94	0.94	0.94
Weighted Average	0.98	0.98	0.98

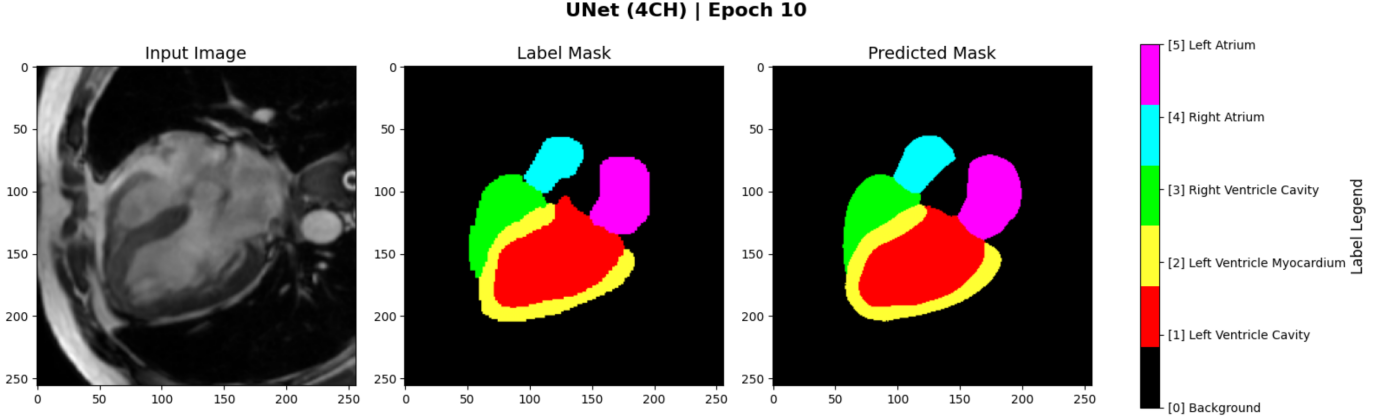


Fig. 22. Additional qualitative visualization results obtained from the proposed U-Net model on the 4CH cardiac MRI dataset. The figure presents representative MRI slices together with their corresponding predicted segmentation masks.

TABLE X
TRAINING, VALIDATION, AND TESTING PERFORMANCE ON THE 2CH DATASET

Epoch	Loss			Dice Score			Accuracy		
	Train	Validation	Test	Train	Validation	Test	Train	Validation	Test
1	0.3018	0.1730	0.1685	0.8770	0.9315	0.9336	0.8770	0.9315	0.9336
5	0.0598	0.0609	0.0598	0.9753	0.9747	0.9751	0.9753	0.9747	0.9751
10	0.0426	0.0488	0.0483	0.9820	0.9797	0.9799	0.9820	0.9797	0.9799

model achieved strong and stable segmentation performance on the 2CH cardiac MRI dataset. At the beginning of training, the network achieved a Dice score of 0.9336 on the testing dataset. As the training process progressed, the segmentation performance improved significantly, reaching a Dice score of 0.9751 at epoch 5 and achieving the best performance at epoch 10 with a Dice score of 0.9799.

In addition, the segmentation loss gradually decreased from 0.1685 to 0.0483, indicating that the model successfully learned meaningful anatomical and spatial representations from the cardiac MRI images.

Figure 23 illustrates the training curves obtained during the optimization process. The figure presents the evolution of segmentation loss, Dice score, and segmentation accuracy across the training epochs.

As shown in Figure 23, the proposed model exhibited stable convergence behavior during the training process. The Dice score and segmentation accuracy increased consistently across the epochs, while the segmentation loss continuously decreased. Furthermore, the close agreement between validation and testing performance indicates good generalization

capability without significant overfitting.

To further evaluate segmentation quality across different cardiac structures, precision, recall, and F1-score metrics were calculated for each segmentation class. Table XI summarizes the obtained per-class segmentation performance on the testing dataset.

The obtained per-class evaluation results demonstrate that the proposed U-Net framework achieved reliable segmentation performance across different anatomical regions within the 2CH cardiac MRI images. Most classes achieved high precision and recall values, indicating accurate feature extraction and effective localization of cardiac boundaries.

The segmentation results also indicate that the model was capable of preserving important anatomical structures while maintaining strong overall segmentation consistency. Although some classes achieved slightly lower performance compared with the background class, the overall results still demonstrate strong segmentation capability and robust model generalization.

Figure 24 presents qualitative segmentation examples generated by the proposed U-Net model on the 2CH dataset.

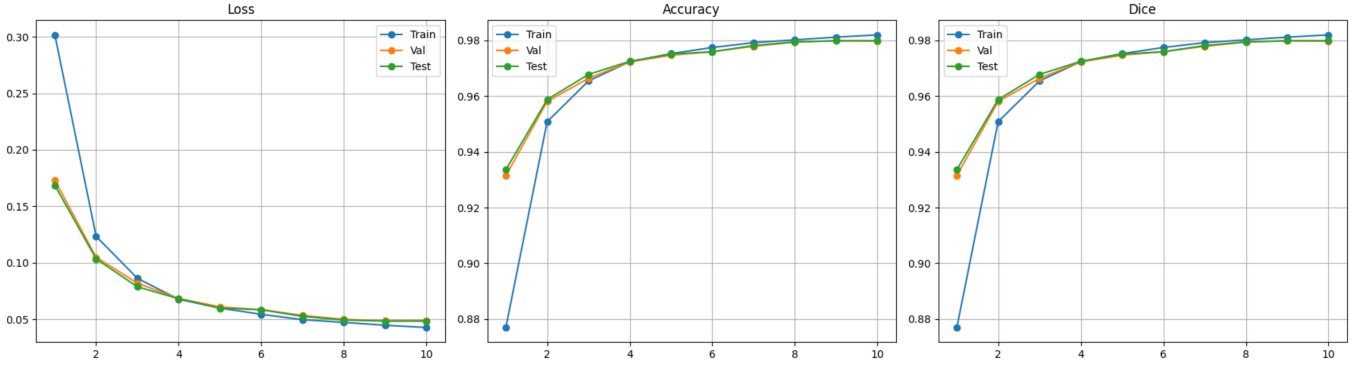


Fig. 23. Training performance curves of the proposed U-Net model on the 2CH cardiac MRI dataset. The figure illustrates the improvement in Dice score and segmentation accuracy together with the gradual reduction in segmentation loss during training.

TABLE XI
PER-CLASS SEGMENTATION PERFORMANCE ON THE 2CH DATASET

Class	Precision	Recall	F1-Score
Background	0.99	0.99	0.99
Left Ventricle Cavity	0.96	0.92	0.94
Left Ventricle Myocardium	0.87	0.90	0.89
Macro Average	0.94	0.94	0.94
Weighted Average	0.98	0.98	0.98

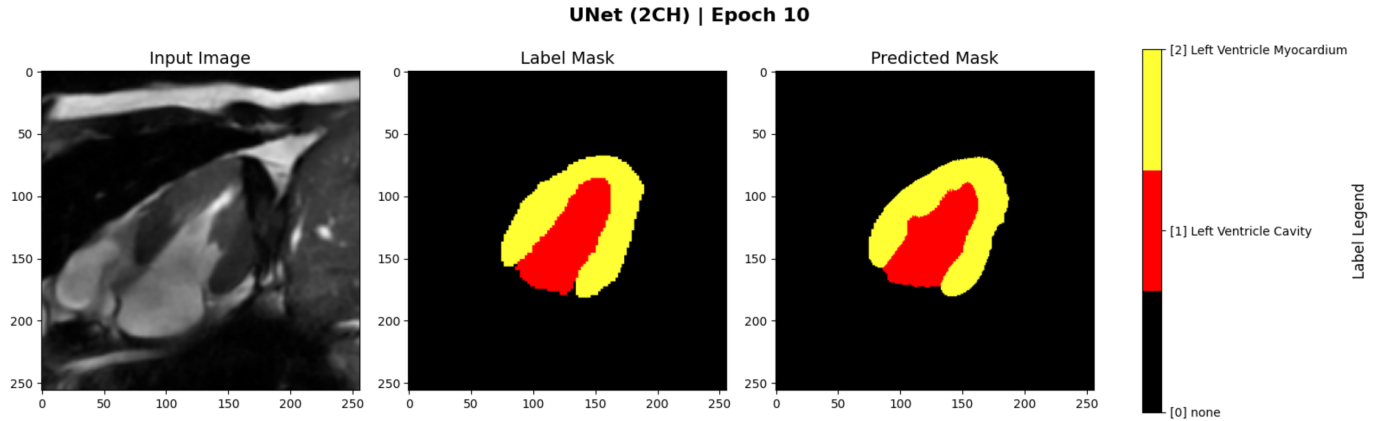


Fig. 24. Qualitative visualization results obtained from the proposed U-Net model on the 2CH cardiac MRI dataset. The figure presents representative MRI slices together with their corresponding predicted segmentation masks.

The figure compares the original MRI slices, corresponding ground-truth masks, and the predicted segmentation outputs generated by the model.

The qualitative visualization results demonstrate strong agreement between the predicted segmentation masks and the corresponding ground-truth annotations. The proposed U-Net model successfully segmented the main cardiac structures while preserving clear anatomical boundaries across different MRI samples.

3) *Evaluation on SAX Cardiac MRI Dataset:* This subsection presents the experimental evaluation of the proposed U-Net architecture on the SAX cardiac cine MRI dataset. The purpose of this experiment was to evaluate the segmentation capability of the proposed framework on short-axis cardiac

MRI images, which are commonly used for ventricular and myocardial analysis in clinical cardiac assessment.

The model was trained for 10 epochs using the same preprocessing pipeline, optimization settings, and experimental configuration applied in the previous experiments.

To summarize the learning behavior of the proposed U-Net framework during training, selected epochs were used for performance comparison. Table XII presents the obtained loss, Dice score, and segmentation accuracy values at epochs 1, 5, and 10.

The obtained experimental results demonstrate that the proposed U-Net architecture achieved excellent segmentation performance on the SAX cardiac MRI dataset. At the beginning of training, the model achieved a Dice score of 0.9728

TABLE XII
TRAINING, VALIDATION, AND TESTING PERFORMANCE ON THE SAX DATASET

Epoch	Loss			Dice Score			Accuracy		
	Train	Validation	Test	Train	Validation	Test	Train	Validation	Test
1	0.1641	0.0731	0.0724	0.9419	0.9725	0.9728	0.9419	0.9725	0.9728
5	0.0311	0.0323	0.0320	0.9873	0.9869	0.9870	0.9873	0.9869	0.9870
10	0.0221	0.0245	0.0249	0.9908	0.9901	0.9900	0.9908	0.9901	0.9900

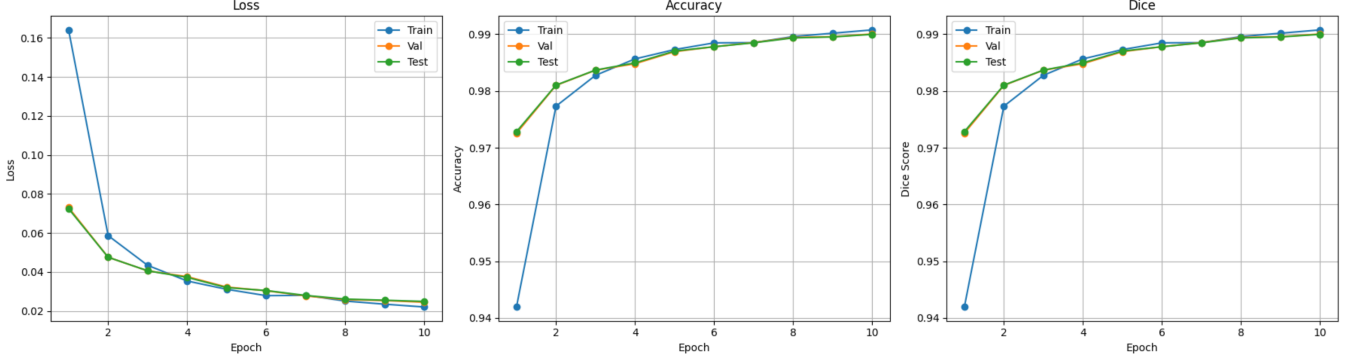


Fig. 25. Training performance curves of the proposed U-Net model on the SAX cardiac MRI dataset. The figure illustrates the improvement in Dice score and segmentation accuracy together with the gradual reduction in segmentation loss during the training process.

TABLE XIII
PER-CLASS SEGMENTATION PERFORMANCE ON THE SAX DATASET

Class	Precision	Recall	F1-Score
Background	1.00	1.00	1.00
Left Ventricle Myocardium	0.93	0.94	0.93
Left Ventricle Cavity	0.98	0.95	0.96
Right Ventricle Cavity	0.95	0.94	0.95
Macro Average	0.96	0.96	0.96
Weighted Average	0.99	0.99	0.99

on the testing dataset. As training progressed, segmentation performance improved consistently, reaching a Dice score of 0.9870 at epoch 5 and achieving the best performance at epoch 10 with a Dice score of 0.9900.

At the same time, the segmentation loss gradually decreased from 0.0724 to 0.0249, indicating that the model successfully learned detailed spatial and anatomical representations from the short-axis cardiac MRI images.

Figure 25 illustrates the training curves obtained during the optimization process. The figure presents the evolution of segmentation loss, Dice score, and segmentation accuracy across the training epochs.

As shown in Figure 25, the proposed model demonstrated highly stable convergence behavior during training. The Dice score and segmentation accuracy increased steadily across the epochs, while the segmentation loss continuously decreased. In addition, the close similarity between validation and testing performance indicates strong generalization capability and reliable segmentation performance on unseen MRI images.

To further evaluate segmentation quality across different anatomical structures, precision, recall, and F1-score metrics were calculated for each segmentation class. Table XIII summarizes the obtained per-class segmentation performance on the testing dataset.

The obtained per-class evaluation results demonstrate that the proposed U-Net framework achieved highly accurate segmentation performance across different cardiac anatomical regions within the SAX MRI images. Most classes achieved strong precision, recall, and F1-score values, indicating accurate feature extraction and reliable localization of cardiac structures.

The overall segmentation performance on the SAX dataset was slightly higher compared with the previous experiments, which may be attributed to the clearer anatomical representation and spatial consistency commonly observed in short-axis cardiac MRI views.

Figure 26 presents qualitative visualization examples generated by the proposed U-Net architecture on the SAX cardiac MRI dataset.

The qualitative visualization results demonstrate strong agreement between the predicted segmentation masks and the corresponding cardiac anatomical structures. The proposed U-Net model successfully preserved fine anatomical boundaries and produced accurate segmentation outputs across different short-axis MRI samples.

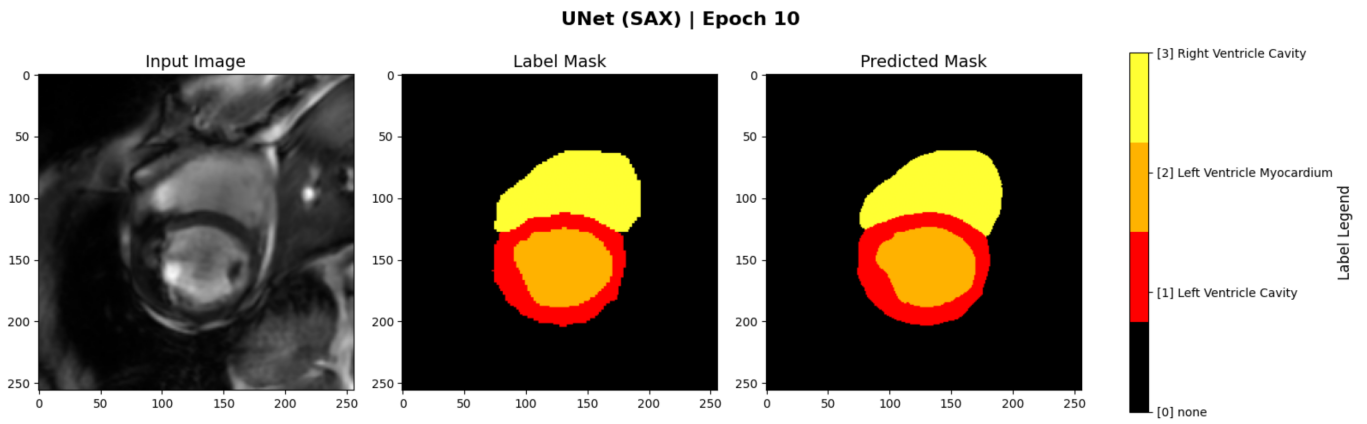


Fig. 26. Qualitative visualization results obtained from the proposed U-Net model on the SAX cardiac MRI dataset. The figure presents representative MRI slices together with their corresponding predicted segmentation masks.

VIII. CONCLUSION

In this work, after performing exploratory data analysis on the provided MICCAI dataset, we concluded that data needs preprocessing for any ML or DL model to work properly and not to get affected by anomalous values in the NIfTI image formats. It is shown that using a local Min-Max scaler gives the best returns while minimizing risks of data poisoning, as any anomalous case will only affect the case itself and will not affect the entire dataset and hence affect the overall reliability of the model. It is also shown that 99 percentile clipping causes some features to get mixed together losing core details which can perplex models. Our work utilized multiple Deep Learning models evaluating each model while tuning parameters and settings to achieve satisfactory results. We have proposed the utilization of the temporal (time) features of the provided NIfTI dataset, as each case is a series of MRI frames together, so, making use of the temporal relationship between frames is shown to increase the segmentation accuracy. This work also proposed the safe usage of data augmentation by incorporating a SAM2 model for automatic segmentation of unlabelled data to increase the portion of the training dataset.

IX. FUTURE WORK AND OPENNESS FOR IMPROVEMENTS

There is an unvisited problem in the CINE CMR images, which is the misalignment of slices due to the motion of the patient. MRIs are slow and the patient usually moves due to continuous breathing, which can cause misalignment between slices. This misalignment can cause imperfections during training or even introduce distorted frames. To address this issue, each frame should have a reference point and adjacent frames should undergo reference mapping. Another suggestion for improvement is the association of information from different views together for the same case. If a patient has a record of 2CH, 4CH and SAX views CMR, we can make use of the fact that all those image views belong to the same patient and try to correlate those views together for a more accurate classification or prediction or even use those different views for depth estimation. Generative AI can also be

used to construct a projected 3D image view of the heart given different CMR views with missing slices where the Generative AI model will predict and estimate those missing slices and construct a 3D instance of the patient's heart.

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